

YAGIZ KOC

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Motivated and passionate computational engineering student specializing in mathematical modeling, simulation, optimization and machine learning, with interest in computational fluid dynamics, uncertainty quantification and decision-making under uncertainty.



SKILLS & INTERESTS

Interest Areas:	Machine Learning & Artificial Intelligence in Engineering, Data Science, Optimization, Simulation Technology, Uncertainty Quantification, Computational Fluid Dynamics (CFD)
Programming Languages:	Python, C++, Julia, MATLAB
Technical Tools:	OpenFOAM, Docker, Git, Microsoft Office

LANGUAGES

German	Full professional proficiency, ÖSD Certificate C1
English	Full professional proficiency, RWTH Language Center Placement Test C2.2.1
Turkish	Native
Bulgarian	Basic proficiency

EXPERIENCE

Oct 2024 – Today	Institute for Combustion Technology (itv) of RWTH Aachen University as Student Assistant - Conducted turbulent fluid simulations for combustion research. - Performed pre- and postprocessing of simulation data to extract meaningful and actionable insights. Bachelor Thesis: “A CFD-based Workflow for Multi-Objective Optimization of Inlet Geometry Under Flashback and NOx Emission Constraints” - Developed a workflow with Bayesian Optimization and evolutionary methods for hydrogen burner shape design that identifies the best balance between minimizing flashback risk for safety and reducing harmful NOx emissions. - Utilized tools: Python, OpenFOAM, Dakota, Git
Oct 2023 – Jun 2024	Institute for Automotive Engineering (ika) of RWTH Aachen University as Student Assistant - Developed functional software for automated and connected driving. - Migrated software components from ROS 1 to ROS 2. - Utilized tools: C++, Docker, ROS 1 and ROS 2, Git
Dec 2022 – Mar 2024	Enevra, Aachen as Technical Team Member - Collaborated with a student team at the start-up Enevra within the Enactus Aachen framework to develop 3D-printed, accessible, and individually adapted prostheses at a fraction of the usual cost and production time, while establishing a sustainable supply structure to improve the quality of life of people in developing countries. - Developed and maintained the software used to customize the prostheses. - Utilized tools: Python, Microsoft Office

EDUCATION

Oct 2022 – Today

RWTH Aachen University

Computational Engineering Science, B.Sc., Grade: 1.3

Tutor positions: "Introduction to Programming" and "Thermodynamics I"

Selection of Taken Courses:

- Machine Learning, Reinforcement Learning, Simulation Technology, Control Systems, Model-based Estimation, Data Analysis and Visualization, Computational Fluid Dynamics, High Performance Computing, Partial Differential Equations, Statistics

Project Thesis: "Systematic Literature Review on Integration of AI Methods in CFD Simulations"

- Conducted within a team of students a systematic literature review to evaluate the current progress and identify research gaps in the integration of AI within computational fluid dynamics workflows.
- Focused personally on turbulence modeling and design optimization with surrogate models.
- Analyzed the role and potential of AI-driven transformation in engineering contexts involving fluid simulations.

Software Engineering Lab Group Project: "Differentiable Programming: Optimizing Non-Differentiable Functions"

- Designed and implemented within a group of students a modular C++ optimization library supporting gradient-based and gradient-free methods (Gradient Descent, Adam, Gradient Sampling, Nelder-Mead).
- Built a comprehensive testing and benchmarking suite using a wide range of test functions to evaluate performance, robustness and convergence.
- Created visualizations to compare the findings.
- Utilized tools: C++, Makefiles, Git

Sep 2017 – Jun 2022

St. Georg Austrian College, Istanbul

Matura (A-levels) Score: 1.0, Grade: 95.23/100, Certificate of Honor

STUDY ABROAD

Feb 2026 – Aug 2026

ETH Zürich

Department of Mechanical and Process Engineering, Exchange Program

Selection of Taken Courses:

- Model Predictive Control, Risk Analysis of Technical Systems, Uncertainty Quantification and Surrogate Modeling, Optimization and Machine Learning, Virtual Reality I: Visualization and Simulation

CERTIFICATIONS & HONORS

Aug 2025

Certificate Career Success in Germany

As part of the FIT for Success @ RWTH Aachen project, the program serves as a proof of special qualifications for entering the professional world in Germany.

Mar 2024 & Mar 2025

Dean's List 2023, 2024 & 2025 at RWTH Aachen University

Nov 2023, Nov 2024

Scholarship Recipient (Deutschlandstipendium)

& Nov 2025

Supported in

- 2023 by Holste-Stiftung
- 2024 by FEV Europe GmbH
- 2025 by NRW.BANK

Jun 2022

Valedictorian of the Class of 2022 at St. Georg Austrian College

Nov 2021

ÖSD Certificate German C1

Grade: 94/100, passed with distinction